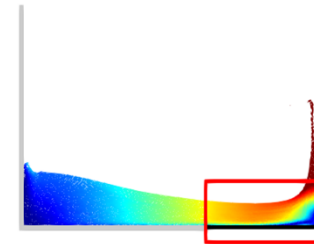


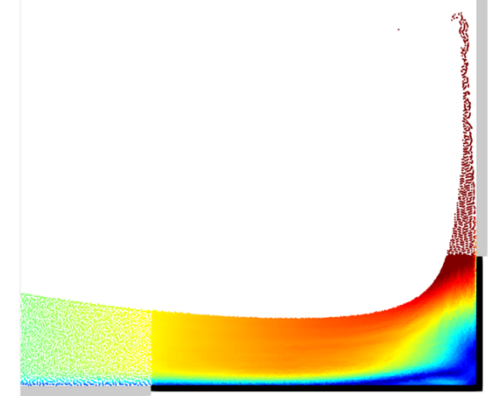
01_DAMBREAK2DVRES

- 2-D dam break flow impacting on the right wall.
One level of refinement is defined in the bottom-right corner, including the refinement of the boundary particles.

[Video](#)

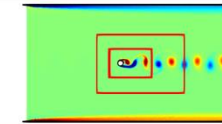


Time: 0.8250 s

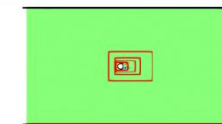
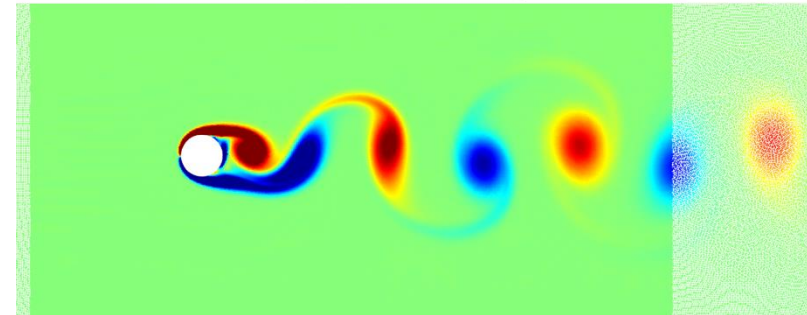


02_FLOWCYLINDERRE

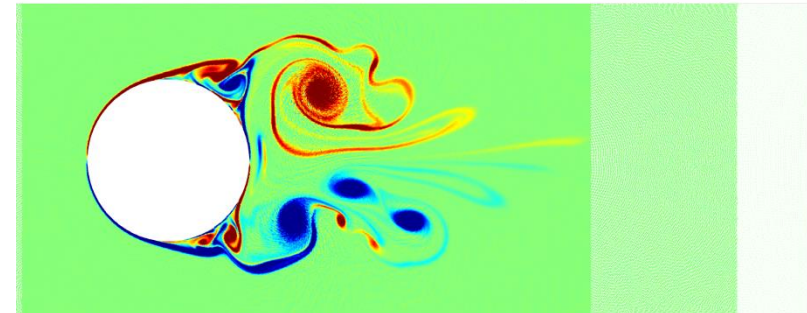
- 2-D flow past a circular cylinder with $Re=200$ ($\nu=0.0005 \text{ m}^2/\text{s}$; $\text{fluidvel}=1 \text{ m/s}$) using 2 levels of refinement. [Video](#)
- 2-D flow past a circular cylinder with $Re=9500$ ($\nu=1.0526 \cdot 10^{-5} \text{ m}^2/\text{s}$; $\text{fluidvel}=1 \text{ m/s}$) using 5 levels of refinement. [Video](#)



Time: 15.00 s



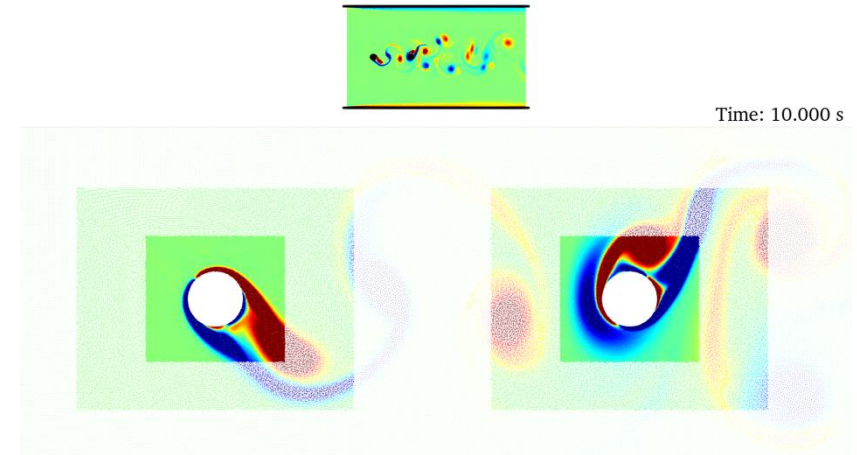
Time: 0.4500 s



03_FLOWCYLINDERTANDEM

- 2-D flow past two circular cylinders with $Re=200$ ($\nu=0.0005 \text{ m}^2/\text{s}$; $fluidvel=1 \text{ m/s}$). Each cylinder oscillates following a sinusoidal movement. 2 levels of refinement are applied to each cylinder, and they follow the movement.

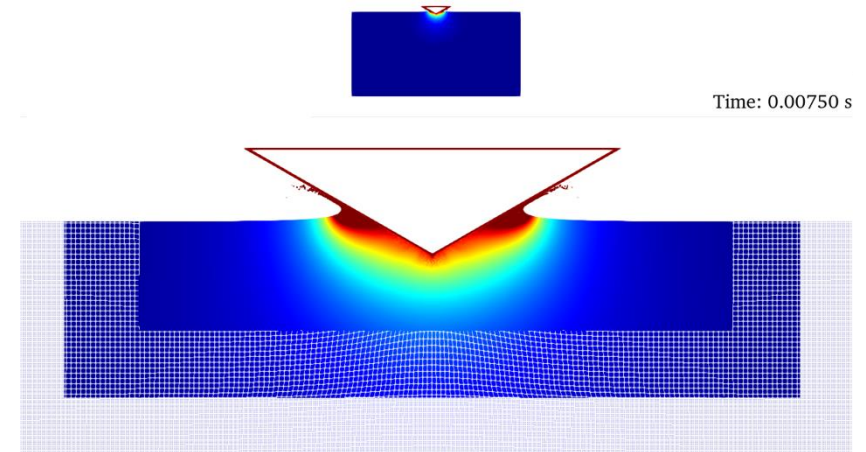
[Video](#)



04_FALLINGWEDGE2D

- 2-D wedge falls with an initial velocity of -6.15 m/s interacting with still water. 2 extra levels of refinement are applied to solve the wedge-fluid interaction, and they follow the movement of the floating wedge.

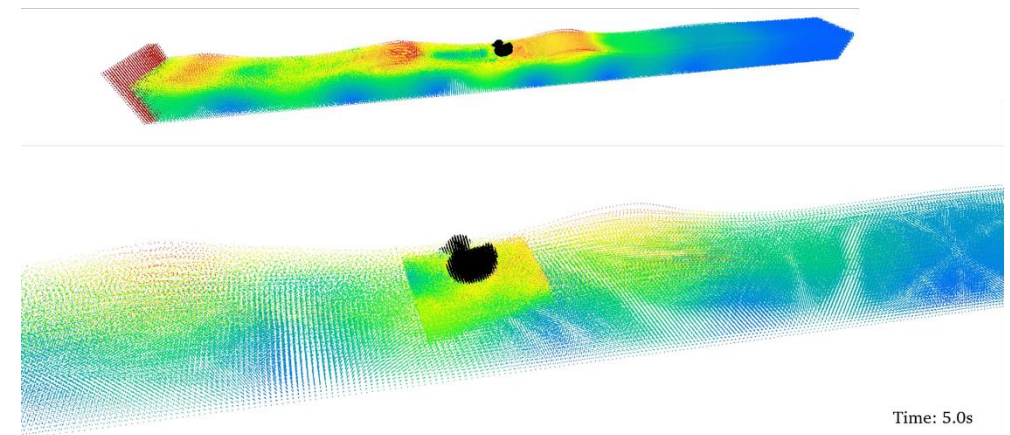
[Video](#)



05_FLOATINGDUCK

- 3-D floating external STL (duck) is simulated under the action of regular waves generated with a piston. One level of refinement is defined around the floating object and follows the movement of the duck.

[Video](#)



06_SPECIAL

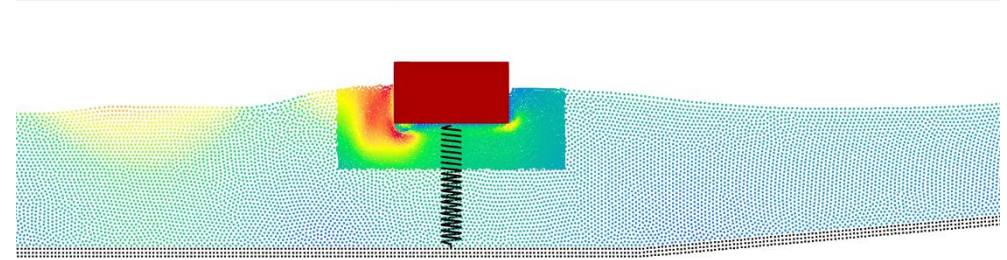
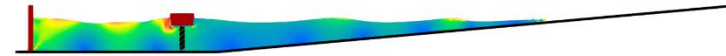
- 2-D floating box under the action of regular waves is restricted to only-heave motion using direct restrictions and simulating the Power Take-Off (PTO) system with a linear spring damper. One level of refinement is defined around the floating box and follows the movement of the box. (CasePointAbsorberSpringVRes)

[Video](#)

- A reproduction of the FSI2 benchmark case from Turek & Hron (2006). The case consists of a 2-D clamped cantilever beam attached to a 2-D cylinder embedded in a channel flow. The Reynolds number is 100, which leads to vortex shedding in the wake of the cylinder. One level of refinement is applied to solve the fluid-structure interaction. (CaseTurekHronFSI2VRes)

[Video](#)

Time: 6.1s



Time: 11.0s

